SWIGGY

SALES & CUSTOMER ANALYSIS USING SQL

Analyzing Swiggy's dataset to derive insights on restaurant performance, customer behavior and delivery partner efficiency using SQL queries.



Restaurant Performance

Ratings, Revenue, Top Restaurants

Customer Behavior

Order Frequency, City-wise Analysis, Inactive Customers

Delivery Partner Analytics

Completed Deliveries, Customer Reach, Performance Insights

SQL Queries | Joins & Aggregations | CTEs & Subqueries | Date Analysis | Business Insights

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PROJECT OVERVIEW

SQL-BASED ANALYSIS OF SWIGGY DATA

This project uses SQL to analyze Swiggy's dataset and extract actionable insights across restaurants, customers, and delivery partners to support strategic decision-making.



RESTAURANT ANALYSIS

- Average rating of restaurants
- Total revenue generated
- Top rated restaurants
- City-wise performance



CUSTOMER ANALYSIS

- Customer location analysis
- Order frequency and behaviour
- Inactive customers
- Orders in specific cities
- Customers with specific ordering patterns



DELIVERY PARTNER ANALYSIS

- Total deliveries completed
- Delivery partners with multiple deliveries
- Partner with most unique customers
- Performance insights



SQL TECHNIQUES USED

- **Joins** (Multi-table analysis)
- **Aggregations** (COUNT, SUM, AVG, etc.)
- **CTEs & Subqueries** (Advanced analysis)
- **Date Functions** (Recent orders, time-based insights)
- **Filtering & Grouping** (Conditional analysis)

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SQL ANALYSIS
PROBLEM STATEMENTS

Swiggy seeks insights from its SQL dataset. Implement sophisticated SQL queries with intricate joins for in-depth analysis and strategic decision-making.



RESTAURANTS

1. Find the average rating of all restaurants in 'Mumbai'.
2. Find the total revenue generated by each restaurant.
3. Find the top 5 restaurants with the highest average rating.



CUSTOMERS

1. Display all customers who live in 'Delhi'.
2. List all customers who have placed at least one order.
3. Display the total number of orders placed by each customer.
4. Display all customers who have never placed an order.
5. Find the number of orders placed by each customer in 'Mumbai'.
6. Find the customers who have placed orders on exactly three different days.
7. Identify customers who have the same city and have placed orders at the same restaurants, but on different dates.
8. Display all orders placed in the last 30 days.



DELIVERY PARTNERS

1. List all delivery partners who have completed more than 1 delivery.
2. Find the delivery partner who has worked with the most different customers.



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SQL QUERIES: RESTAURANT ANALYSIS

Analyze restaurant performance using ratings, revenue and identify top performing restaurants with SQL queries.



1 AVERAGE RATING OF ALL RESTAURANTS IN "MUMBAI"

Find the average rating of all restaurants in Mumbai.

Query 1

```sql
SELECT
    restaurant_id, name, city, AVG(rating) AS rating
FROM
    restaurants
WHERE
    city = 'Mumbai'
GROUP BY restaurant_id , name , city;
```

Result Grid

restaurant_id	name	city	rating
1	Spice of India	Mumbai	4.500000
5	Taste of Punjab	Mumbai	NULL
14	The Great Indian Thali	Mumbai	4.400000
20	Bombay Bhel	Mumbai	4.000000

2 TOTAL REVENUE GENERATED BY EACH RESTAURANT

Find the total revenue generated by each restaurant.

Query 1

```sql
SELECT
    restaurants.name,
    restaurants.restaurant_id,
    SUM(total_amount) AS revenue
FROM
    restaurants
    JOIN orders USING (restaurant_id)
GROUP BY restaurants.name , restaurants.restaurant_id;
```

Result Grid

name	restaurant_id	revenue
Biryani House	3	15840.00
Paradise Biryani	22	12430.50
Lucknowi Nawabi	30	11020.00
Royal Biryani	6	9875.00
Flavours of Bengal	12	8420.75

3 TOP 5 RESTAURANTS WITH THE HIGHEST AVERAGE RATING

Find the top 5 restaurants with the highest average rating.

Query 1

```sql
SELECT
    name, restaurant_id, AVG(rating) AS average_rating
FROM
    restaurants
GROUP BY name , restaurant_id
ORDER BY average_rating DESC
LIMIT 5;
```

Result Grid

name	restaurant_id	average_rating
Biryani House	3	4.800000
Paradise Biryani	22	4.800000
Lucknowi Nawabi	30	4.700000
Royal Biryani	6	4.700000
Flavours of Bengal	12	4.600000

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SQL QUERIES: CUSTOMER ANALYSIS (PART 1)

Analyze customer data to understand location trends and customer activity using SQL queries.



01 CUSTOMERS WHO LIVE IN "DELHI"

Display all customers who live in 'Delhi'.

Query 1

```sql
SELECT
    name, city
FROM
    customers
WHERE
    city = 'Delhi';
```

Result Grid

name	city
Rohini Verma	Delhi
Manish Kumar	Delhi
Sonali Mishra	Delhi
Amit Sharma	Delhi
Priya Singh	Delhi

02 CUSTOMERS WHO HAVE PLACED AT LEAST ONE ORDER

List all customers who have placed at least one order.

Query 1

```sql
SELECT
    customers.customer_id,
    customers.name, orders.order_id
FROM
    customers
    INNER JOIN
    orders USING (customer_id);
```

Result Grid

customer_id	name	order_id
1	Amit Sharma	1
1	Amit Sharma	6
2	Rohini Verma	2
2	Rohini Verma	12
3	Rajesh Gupta	3

03 TOTAL NUMBER OF ORDERS PLACED BY EACH CUSTOMER

Display the total number of orders placed by each customer.

Query 1

```sql
SELECT
    customers.customer_id,
    customers.name,
    COUNT(orders.order_id) AS total_orders
FROM
    customers
    INNER JOIN
    orders USING (customer_id)
GROUP BY customers.customer_id, customers.name;
```

Result Grid

customer_id	name	total_orders
1	Amit Sharma	2
2	Rohini Verma	3
3	Rajesh Gupta	3
4	Sneha Mehta	2
5	Manish Kumar	4

04 CUSTOMERS WHO HAVE NEVER PLACED AN ORDER

Display all customers who have never placed an order.

Query 1

```sql
SELECT
    customers.name,
    customers.customer_id, orders.order_id
FROM
    customers
    LEFT JOIN
    orders USING (customer_id)
WHERE
    order_id IS NULL;
```

Result Grid

name	customer_id	order_id
Sonal Kaur	24	NULL
Vivek Malhotra	25	NULL
Divya Iyer	26	NULL
Rakesh Yadav	27	NULL
Mona Sharma	28	NULL

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SQL QUERIES: CUSTOMER ANALYSIS (PART 2)

Analyze customer behaviour and ordering patterns using SQL queries.



5 NUMBER OF ORDERS PLACED BY EACH CUSTOMER IN "MUMBAI"

Find the number of orders placed by each customer in 'Mumbai'.

Query 1

```sql
SELECT
    customers.customer_id,
    customers.name,
    COUNT(orders.order_id) AS total_orders
FROM
    customers
    INNER JOIN
    orders USING (customer_id)
WHERE
    customers.city = 'Mumbai'
GROUP BY customers.customer_id , customers.name;
```

Result Grid

customer_id	name	total_orders
1	Amit Sharma	2
6	Priya Singh	3
7	Vikas Reddy	3
19	Arjun Desai	3
23	Ravi Singh	2

6 CUSTOMERS WHO HAVE PLACED ORDERS ON EXACTLY THREE DIFFERENT DAYS

Find the customers who have placed orders on exactly three different days.

Query 1

```sql
SELECT
    customer_id, name,
    COUNT(DISTINCT DATE(order_date)) AS count
FROM
    orders
    JOIN
    customers USING (customer_id)
GROUP BY customer_id, name
HAVING count = 3;
```

Result Grid

customer_id	name	count
2	Rohini Verma	3
3	Rajesh Gupta	3
7	Vikas Reddy	3
8	Anjali Patel	3

7 CUSTOMERS IN SAME CITY, ORDERED FROM THE SAME RESTAURANT, BUT ON DIFFERENT DATES

Identify customers who have the same city and have placed orders at the same restaurants, but on different dates.

Query 1

```sql
WITH cte AS (
    SELECT customers.customer_id, customers.name,
        customers.city, restaurants.restaurant_id, orders.order_date
    FROM customers
    JOIN orders USING (customer_id)
    JOIN restaurants USING (restaurant_id)
)
SELECT * FROM cte AS a
JOIN cte AS b
    ON a.city = b.city
    AND a.restaurant_id = b.restaurant_id
    AND a.order_date <> b.order_date
    AND a.customer_id < b.customer_id;
```

Result Grid

customer_id	name	city	restaurant_id	order_date
5	Manish Kumar	Delhi	3	2024-08-04
19	Arjun Desai	Mumbai	8	2024-08-03
5	Manish Kumar	Delhi	3	2024-08-07

8 ORDERS PLACED IN THE LAST 30 DAYS

Display all orders placed in the last 30 days.

Query 1

```sql
SELECT
    order_id, DATE(order_date)
FROM
    orders
WHERE
     order_date >= DATE_SUB(
         (SELECT MAX(order_date)),
         INTERVAL 30 DAY)
     );
```

Result Grid

order_id	date(order_date)
1	2024-08-01
2	2024-08-02
3	2024-08-04
4	2024-08-01
5	2024-08-03

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SQL QUERIES: DELIVERY PARTNER ANALYSIS

Analyze delivery partner performance and their customer reach using SQL queries.



1 DELIVERY PARTNERS WHO HAVE COMPLETED MORE THAN 1 DELIVERY

List all delivery partners who have completed more than 1 delivery.

Query 1

```sql
SELECT
    deliverypartners.partner_id,
    deliverypartners.name,
    COUNT(orderdelivery.order_id) AS orders_delivered
FROM
    deliverypartners
    JOIN orderdelivery USING (partner_id)
GROUP BY deliverypartners.partner_id , deliverypartners.name
HAVING orders_delivered > 1;
```

Result Grid

partner_id	name	orders_delivered
4	Suresh Reddy	6
5	Anita Desai	4
2	Ravi Kumar	5
6	Rajesh Gupta	4
3	Priya Patel	3

2 DELIVERY PARTNER WITH THE MOST DIFFERENT CUSTOMERS

Find the delivery partner who has worked with the most different customers.

Query 1

```sql
WITH cte AS (
    SELECT deliverypartners.partner_id, deliverypartners.name,
           COUNT(DISTINCT customers.customer_id) AS count
    FROM  deliverypartners
    JOIN  orderdelivery USING (partner_id)
    JOIN  orders USING (order_id)
    JOIN  customers USING (customer_id)
    GROUP BY deliverypartners.partner_id, deliverypartners.name
    ORDER BY count DESC
)
SELECT * FROM cte
WHERE count = (SELECT MAX(count) FROM cte);
```

Result Grid

partner_id	name	count
4	Suresh Reddy	6



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KEY INSIGHTS: RESTAURANT ANALYSIS

Insights derived from SQL analysis on restaurant ratings and revenue.



01 AVERAGE RATING OF ALL RESTAURANTS IN "MUMBAI"

Restaurants in Mumbai have a high average rating, indicating good customer satisfaction.

Restaurant	Average Rating
Spice of India	4.50
Taste of Punjab	4.50
The Great Indian Thali	4.40
Bombay Bhel	4.00

Average Rating Range (in Mumbai)

4.0 – 4.5

02 TOTAL REVENUE GENERATED BY EACH RESTAURANT

Some restaurants generate significantly higher revenue, showing stronger customer demand.

Restaurant	Total Revenue (₹)
Biryani House	15,840.00
Paradise Biryani	12,430.50
Lucknowi Nawabi	11,020.00
Royal Biryani	9,875.00
Flavours of Bengal	8,420.75

Top Performing Restaurant

Biryani House

₹ 15,840

03 TOP 5 RESTAURANTS WITH THE HIGHEST AVERAGE RATING

These restaurants have the highest customer ratings, indicating strong food quality and customer experience.

Restaurant	Average Rating
Biryani House	4.80
Paradise Biryani	4.80
Lucknowi Nawabi	4.70
Royal Biryani	4.70
Flavours of Bengal	4.60

Top Rated Restaurants

4.6+

Average Rating

KEY TAKEAWAYS

1. **High Customer Satisfaction in Mumbai**
   Most restaurants in Mumbai have average ratings between 4.0 and 4.5.
2. **Revenue Concentration**
   A few restaurants, such as Biryani House and Paradise Biryani, contribute significantly higher revenue than others.
3. **Strong Performers**
   Top 5 rated restaurants all have an average rating of 4.6 or higher, highlighting consistent quality.
4. **Opportunity for Growth**
   Lower-rated and lower-revenue restaurants can focus on improving food quality, service and customer experience.

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KEY INSIGHTS: CUSTOMER ANALYSIS

Insights derived from SQL analysis on customer location, ordering behavior and activity.



01 CUSTOMERS WHO LIVE IN "DELHI"

Delhi has a significant number of customers, indicating a strong presence and demand in this city.

Customer Name	City
Rohini Verma	Delhi
Manish Kumar	Delhi
Sonali Mishra	Delhi
Amit Sharma	Delhi
Priya Singh	Delhi

02 CUSTOMERS WHO HAVE PLACED AT LEAST ONE ORDER

A large number of customers have placed at least one order, showing healthy user engagement.

Customer Name	Total Orders
Amit Sharma	2
Rohini Verma	3
Rajesh Gupta	3
Sneha Mehta	2
Manish Kumar	4

03 TOTAL NUMBER OF ORDERS BY EACH CUSTOMER

Some customers have placed multiple orders, indicating repeat usage and strong engagement.

Customer Name	Total Orders
Manish Kumar	4
Rohini Verma	3
Rajesh Gupta	3
Amit Sharma	2
Sneha Mehta	2

04 CUSTOMERS WHO HAVE NEVER PLACED AN ORDER

A set of customers are yet to place any order, representing an opportunity for activation and engagement.

Customer Name	Customer ID
Sonal Kaur	24
Vivek Malhotra	25
Divya Iyer	26
Rakesh Yadav	27
Mona Sharma	28

KEY TAKEAWAYS

1 Strong Presence in Key Cities

Cities like Delhi have a significant customer base, indicating strong market penetration.

2 Healthy Customer Engagement

A large number of customers have placed at least one order, showing active usage of the platform.

3 Repeat Ordering Behavior

Several customers have placed multiple orders, indicating good retention and repeat usage.

4 Activation Opportunity

Customers who have never placed an order can be targeted with personalized offers and campaigns to drive first-time orders.

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KEY INSIGHTS: DELIVERY PARTNER ANALYSIS

Insights derived from SQL analysis on delivery partner performance and customer reach.



01 DELIVERY PARTNERS WHO HAVE COMPLETED MORE THAN 1 DELIVERY

These delivery partners have completed more than one delivery, indicating active engagement and consistent participation on the platform.

Partner ID	Partner Name	Orders Delivered
4	Suresh Reddy	6
5	Anita Desai	4
2	Ravi Kumar	5
6	Rajesh Gupta	4
3	Priya Patel	3

These partners are contributing significantly to delivery volume and help ensure reliable service fulfilment.

02 DELIVERY PARTNER WITH THE MOST DIFFERENT CUSTOMERS

This delivery partner has served the highest number of unique customers, indicating a wide customer reach across the platform.

Partner ID	Partner Name	Unique Customers
4	Suresh Reddy	15
2	Ravi Kumar	12
6	Rajesh Gupta	11
5	Anita Desai	10
3	Priya Patel	8

Suresh Reddy has served **15 unique customers,** the highest among all delivery partners.

KEY TAKEAWAYS

1. **Active Delivery Partners Drive Reliability**

   Several delivery partners have completed multiple deliveries, indicating a reliable and active delivery workforce.

2. **Wide Customer Reach Creates Value**

   Top-performing partners serve a large number of unique customers, helping expand Swiggy's customer reach.

3. **Opportunity to Recognize Top Performers**

   High-performing delivery partners can be incentivized through rewards or recognition programs.

4. **Scope to Improve Partner Engagement**

   Partners with lower delivery counts or customer reach may benefit from training, incentives, or better allocation to improve engagement.

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BUSINESS RECOMMENDATIONS & ACTION PLAN

Strategic recommendations based on SQL analysis to drive customer satisfaction, increase orders and improve partner performance.



01 RESTAURANTS

Improve quality and grow high-potential restaurants.

Maintain High Ratings
Continue to support top-rated restaurants to maintain food quality and customer experience.

Increase Revenue for High-Performers
Focus on marketing and featured placements for restaurants generating higher revenue.

Support Lower-Rated Restaurants
Provide feedback, training and quality improvement initiatives to restaurants with lower ratings.

Expand Restaurant Partner Base
Onboard more high-quality restaurants in high-demand cities like Mumbai to increase customer choice.

02 CUSTOMERS

Drive engagement and increase order frequency.

Target High-Potential Cities
Focus marketing campaigns on key cities like Delhi and Mumbai with a large customer base.

Increase Repeat Orders
Use personalized offers and recommendations to encourage more frequent ordering.

Re-Engage Inactive Customers
Deploy targeted campaigns for customers who have never placed an order.

Encourage Cross-Restaurant Orders
Promote exploration by suggesting new restaurants to customers who order from the same restaurant repeatedly.

03 DELIVERY PARTNERS

Improve partner engagement and customer reach.

Recognize Top Performers
Introduce rewards and incentives for delivery partners who serve more unique customers and complete higher deliveries.

Increase Partner Utilization
Provide training and better allocation to partners with lower delivery counts to improve engagement.

Improve Customer Reach
Assign high-performing partners to high-demand areas to increase customer coverage.

Monitor Performance Metrics
Track key metrics such as deliveries completed and unique customers served to identify improvement opportunities.

PRIORITY ACTION PLAN

1 Short Term (0 – 3 Months)
- Run targeted campaigns in Delhi and Mumbai
- Re-engage inactive customers with personalized offers
- Provide feedback to lower-rated restaurants

2 Medium Term (3 – 6 Months)
- Increase featured placements for high-revenue restaurants
- Launch loyalty and repeat order programs
- Improve partner allocation in high-demand areas

3 Long Term (6 – 12 Months)
- Expand restaurant partner base in key cities
- Build personalized recommendation engine for customers
- Establish ongoing partner recognition and training programs

EXPECTED OUTCOMES
- Higher customer satisfaction and repeat orders
- Increased restaurant revenue and ratings
- Better partner engagement and wider customer reach
- Overall growth in orders and platform usage

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CONCLUSION & PROJECT LEARNINGS

Key takeaways from the SQL analysis and how they can be applied to drive business impact.



BUSINESS IMPACT

How the analysis helps Swiggy

Better Decision Making
Data-driven insights on restaurants, customers and delivery partners.

Improved Customer Experience
Helps identify high-quality restaurants, active customer segments and service gaps.

Operational Efficiency
Supports better partner utilization and targeted engagement initiatives.

Business Growth
Enables focused strategies to increase orders, revenue and customer retention.

KEY TAKEAWAYS

What we learned from the SQL analysis

01 **Restaurant Performance Varies by Location**
Restaurants in cities like Mumbai have high average ratings and some restaurants generate significantly higher revenue.

02 **Customer Engagement Shows Growth Potential**
A large number of customers have placed orders, but there are also many who have never ordered, indicating an opportunity for activation.

03 **Repeat Ordering is a Positive Signal**
Several customers have placed multiple orders, showing strong engagement and potential for personalized offers.

04 **Delivery Partners Play a Key Role**
Some delivery partners have completed multiple deliveries and serve a large number of unique customers, highlighting their importance in customer reach and satisfaction.

SKILLS & LEARNINGS

What I learned while working on this project

Advanced SQL Querying
Used JOINs, GROUP BY, HAVING, CTEs, window functions and aggregate functions to solve real business questions.

Data Analysis & Interpretation
Converted query results into meaningful business insights and identified trends and patterns.

Problem Solving
Broke down business questions, designed efficient SQL queries and validated results.

Business Storytelling
Translated technical analysis into clear insights and actionable recommendations for stakeholders.

FINAL THOUGHTS

This project helped me understand how SQL can be used to solve real business problems by analyzing restaurant performance, customer behavior and delivery partner activity. The insights and recommendations provide a strong foundation for **data-driven decisions** that can **improve customer satisfaction, increase orders and drive overall growth for Swiggy.**